Residual Cohomology of Free Pure-Weyl Gravity on the Conformal Cylinder: Derived Reduction, Weyl-Square Classes, and Krein Completion

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Abstract

We compute the absolute residual $SO(4,2)$ cohomology of free four-dimensional pure-Weyl gravity on $\mathbb{R} \times S^3$ in a selected closed-universe BV–BFV polarization. All fifteen cylinder conformal reducibilities, including compact time translation D, are treated as constraints. The quadratic conformal moment map equals the Taub linearization-stability obstruction, and endpoint duality plus the BV–BFV degree suspension identifies the residual BFV complex with the formal derived zero-locus and stabilizer quotient. Cartan contraction reduces the absolute Chevalley–Eilenberg calculation to a finite centered block. Its vacuum and one-particle sectors are acyclic, while

$$H^4_{\mathrm{res}} = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad \mathrm{sig}\, G_{\mathrm{res}} = (2,0), \qquad G_{\mathrm{res}} = I_2$$

after the stated basis, orientation, and unit-overlap normalization. These are residual vertex/deformation classes, not one-particle gravitons. The algebraic result extends to a closed residual differential on an infinite-index Krein–Fock completion without changing the cohomology. Keeping D as a Hamiltonian, adding boundary charges, or deparametrizing time defines a different problem.

1 Problem, selection, and claim boundary

Let

$$M = \mathbb{R} \times S^3, \qquad \bar{g} = -dt^2 + d\Omega_3^2, \tag{1}$$

and consider the linearized pure-Weyl theory about $\bar{g}$. The local Diff×Weyl complex has fifteen global reducibilities: the conformal Killing transformations of the cylinder. There are two inequivalent ways to treat them. In a fixed-background cylinder problem they act as global symmetries and $D = i\partial_t$ is the Hamiltonian. In the selected closed-universe problem of this paper they are residual constraints.

We assume throughout that S^3 is closed, no boundary or asymptotic charge is retained, no external clock is adjoined, the cylinder is not deparametrized, and all fifteen residual reducibilities are gauged. The derived statement is formal near $\bar{g}$ and concerns the zero-charge, second-order

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integrable sector. We use the infinitesimal stabilizer algebra $\mathfrak{g} \cong \mathfrak{so}(4,2)$; no integrated group representation or free group action is assumed.

The output must also be typed correctly. The local BV differential, current, moment map, and endpoint pairing are classical linearized data. The symmetric-algebra coefficient module arises only after a positive-frequency Lagrangian polarization. Its Hilbert/Krein completion is a selected free state representation, not the unpolarized classical BV tangent complex. Positivity below belongs only to two ghost-dressed residual deformation classes and is not a positive graviton norm.

2 The parity-complete Weyl module

The on-shell chiral Weyl module has character

$$\chi_+(q) = \frac{5q^2 - 9q^4 + 4q^5}{(1-q)^4}, \tag{2}$$

and decomposes on the cylinder into three multiplicity-free towers

$$\begin{aligned}
E_n^+ &: \left(\frac{n+2}{2}, \frac{n-2}{2}\right), & \dim E_n &= (n+3)(n-1), & n &\geq 2,\\
A_n^+ &: \left(\frac{n}{2}, \frac{n-2}{2}\right), & \dim A_n &= (n+1)(n-1), & n &\geq 3,\\
L_n^+ &: \left(\frac{n}{2}, \frac{n-4}{2}\right), & \dim L_n &= (n+1)(n-3), & n &\geq 4.
\end{aligned} \tag{3}$$

The negative chirality exchanges the two $SU(2)$ spins. The smooth BGG bridge and explicit all-level cylinder intertwiners identify this module with the parity-complete geometric Weyl-curvature quotient. This paper uses that theorem as the one-particle input; its coordinate certificates are listed in Section 7.

The invariant form is nondegenerate and indefinite:

$$J_{\text{conf}} = +I_E \oplus (-I_A) \oplus (-I_L). \tag{4}$$

Proper conformal generators obey $(K_q^-)^\sharp = K_q^+$ with respect to J_{conf} and mix the three towers. Hence their signs cannot be changed independently. The polarized algebraic coefficient module is

$$\mathcal{F}_{\mathcal{W}} = \mathrm{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-). \tag{5}$$

This is the characteristic-zero symmetric-algebra lift of the free linearized BRST cohomology, not an additional classical field complex.

3 Taub constraints and the derived residual quotient

For a residual reducibility $\epsilon = (\xi, \sigma)$, with $2\nabla_{(\mu}\xi_{\nu)} + 2\sigma\bar{g}_{\mu\nu} = 0$, let h solve the linearized Bach equation. The second-order Bach equation is

$$B^{(1)}[k] + B^{(2)}[h,h] = 0. \tag{6}$$

Proposition 3.1 (Taub equals moment map)**.** *The current $J_\epsilon^\mu = \xi_\nu B^{(2)\mu\nu}[h,h]$ is conserved. If h is tangent to a second-order family of Bach-flat metrics, then*

$$\mathcal{T}_\epsilon[h] = \int_{S^3} n_\mu \xi_\nu B^{(2)\mu\nu}[h,h]\, d\Sigma = 0. \tag{7}$$

With the action and orientation conventions used here, this charge is the quadratic Hamiltonian moment map:

$$\mu_\epsilon(h) = \frac{1}{2}\Omega_\Sigma(h, R_\epsilon h) = \mathcal{T}_\epsilon[h]. \tag{8}$$

Proof. The second-order divergence and trace identities give $\nabla_\mu B^{(2)\mu\nu} = 0$ and $B^{(2)\mu}{}_\mu = 0$ on the linearized shell. Contracting with the conformal Killing equation proves current conservation. If (6) is solvable, the charge is minus the integrated linearized Noether current of k, which is a spatial divergence for a reducibility parameter and vanishes because $\partial S^3 = \varnothing$. The covariant phase-space Noether identity gives the first equality in (8); direct D normalization and conformal equivariance fix its scalar for all fifteen components [1, 2]. □

The endpoint module is not introduced by hand. In the finite-dimensional residual zero-mode block of the cyclic minimal detour chain, write

$$G \xrightarrow{A} M \xrightarrow{B} E \xrightarrow{C} I, \qquad C = \pm A^\sharp, \tag{9}$$

and set $Z = \ker A \cong \mathfrak{g}$.

Proposition 3.2 (Endpoint duality and suspension)**.** *There is a canonical perfect pairing*

$$\Theta : I/\operatorname{im} C \xrightarrow{\sim} Z^*, \qquad \Theta([u])(z) = \langle z, u\rangle_{G,I}. \tag{10}$$

The selected algebraic time-slice BV–BFV map suspends this quotient by one degree,

$$\tau : H^{\text{bulk}}_{\text{endpoint}} \longrightarrow Z^*[-1]_{\text{BFV}}, \qquad \tau = \Theta \tag{11}$$

in the selected canonical orientation. Thus exactly one residual cotangent pair (c^a, b_a) of degrees $(+1, -1)$ is retained.

Proof. For $z \in \ker A$, adjointness gives $\langle z, Ce\rangle = 0$, so Θ is well defined. Its kernel vanishes because

$$(\ker A)^\perp = \operatorname{im} A^\sharp = \operatorname{im} C.$$

The fifteen-dimensional endpoint and kernel dimensions give surjectivity. Independently of this cokernel argument, the selected algebraic time-slice construction gives an equivariant map $\tau = \lambda\Theta$. Choose representatives $\Theta([u_a]) = e^a$. The separately certified endpoint/Taub obstruction map gives $Q_{\text{bulk}} u_a = \mu_a$ in the quotient, while the canonical ghost form $\Omega_{\text{gh}} = \delta b_a \wedge \delta c^a$ and the BFV charge (12) give $Q_{\text{BFV}} b_a = \mu_a$ on the ghost-free slice. Compatibility of the time-slice chain map therefore fixes $\lambda = +1$; equivariance extends the independently normalized D component to all fifteen components. Thus endpoint duality, the Taub map, and the suspension normalization are three successive inputs, not definitions of one another. □

Let $\mathcal{P}_{\text{lin}}$ be the formal linearized phase space after the local gauge quotient. The selected residual BFV charge is

$$\Omega_{\text{res}} = c^a \mu_a - \frac{1}{2} f^a{}_{bc} c^b c^c b_a. \tag{12}$$

Theorem 3.3 (Derived residual reduction)**.** *Under the assumptions of Section 1, the residual classical problem is represented in the selected formal algebraic category by*

$$\boxed{\mathcal{P}_{\mathrm{der}} = \left[\mathcal{P}_{\mathrm{lin}} \overset{\mathbf{R}}{\underset{\mathfrak{g}^*}{\times}} \{0\} /\!/^{\mathbf{R}} \mathfrak{g}\right].} \tag{13}$$

Equivalently, (12) *generates*

$$Qf = c^a\{\mu_a, f\}, \qquad Qc^a = -\frac{1}{2} f^a{}_{bc} c^b c^c, \qquad Qb_a\big|_{c=0} = \mu_a. \tag{14}$$

After positive-frequency polarization this becomes the absolute residual Chevalley–Eilenberg complex on (5)*.*

Proof. Proposition 3.1 identifies the equations imposed by the Koszul part of (14) with the actual quadratic Taub obstruction components; in particular, every second-order integrable perturbation lies in their common zero locus. No converse sufficiency claim is used here. Proposition 3.2 identifies its equation values with $\mathfrak{g}^*$, supplies the degree--1 BFV momenta, and prevents a duplicate endpoint sector. The canonical brackets applied to (12) give (14); equivariance of μ and the Jacobi identity give $Q^2 = 0$. Its Koszul part resolves the formal derived zero fibre and its Chevalley–Eilenberg part takes the infinitesimal stabilizer quotient, proving (13). This retains nonregular stabilizer directions rather than replacing them by an ordinary smooth quotient [3]. In the chosen polarization c^a acts by exterior multiplication and b_a by contraction, producing the absolute complex below. □

Remark 3.4 (Fixed-cylinder alternative). If D is a Hamiltonian or the moment-map values are boundary charges, one does not impose $\mu = 0$ or divide by the full stabilizer. The relative primary kernel remains useful representation theory but is not (13). A subgroup, boundary, or deparametrized problem requires its own relative or boundary BFV complex; deleting only the D ghost is not consistent because $[K^-, K^+]$ contains D.

4 Cartan contraction and the exact residual group

The absolute complex is

$$C^q = \mathcal{F}_{\mathcal{W}} \otimes \Lambda^q \mathfrak{g}^*, \qquad d = c^a \rho(G_a) - \frac{1}{2} f^a{}_{bc} c^b c^c \iota_a. \tag{15}$$

The compact grading is

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \qquad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4), \tag{16}$$

and total compact degree is $\delta = D_{\mathrm{matter}} + D_{\mathrm{ghost}}$. The centered ghost vacuum v_- is the determinant of the four ghosts dual to $\mathfrak{g}_{+1}$, so

$$\mathrm{gh}(v_-) = 4, \qquad D_{\mathrm{ghost}}(v_-) = -4. \tag{17}$$

Theorem 4.1 (Cartan reduction)**.** *Inclusion of the centered subcomplex is a quasi-isomorphism:*

$$H^\bullet(C, d) \cong H^\bullet(C_{\delta=0}, d). \tag{18}$$

Proof. The Cartan identity is $d\iota_D + \iota_D d = \mathcal{L}_D$. On C_δ with $\delta \neq 0$, $h_\delta = \iota_D/\delta$ satisfies $dh_\delta + h_\delta d = 1$. □

The ghost spectrum is bounded below by -4, whereas every one-particle matter excitation has energy at least two. At ghost number four and total degree zero only particle numbers $N = 0, 1, 2$ occur. The vacuum group $H^4(\mathfrak{g}; \mathbb{C})$ vanishes. For one chirality the relevant one-particle window has dimensions and exact ranks

$$C_0^3(290) \xrightarrow{d_3} C_0^4(1311) \xrightarrow{d_4} C_0^5(3657), \qquad \operatorname{rank} d_3 = 260, \quad \operatorname{rank} d_4 = 1051. \tag{19}$$

Thus the middle cohomology is zero; parity gives the other chirality.

At two particles the minimal energy is four and exterior saturation removes the incoming degree:

$$0 \longrightarrow C_0^4(55) \xrightarrow{d_4} C_0^5(385), \qquad \operatorname{rank} d_4 = 53. \tag{20}$$

The two exact kernel vectors are the normalized same-chirality singlets

$$|W_\pm^2\rangle = \sqrt{\frac{2}{5}}|2,-2\rangle_\pm - \sqrt{\frac{2}{5}}|1,-1\rangle_\pm + \frac{1}{\sqrt{5}}|0,0\rangle_\pm. \tag{21}$$

The rank 53 is verified both by the authoritative absolute CE calculation and by an independent integral spin-two construction over two good primes.

For the pairing, orient determinant vectors $v_- \in \det \mathfrak{g}_{+1}^*$, $\Theta_0 \in \det \mathfrak{g}_0^*$, and $v_+ = v_-^\dagger \in \det \mathfrak{g}_{-1}^*$. Coefficient extraction from $v_+ \wedge \Theta_0 \wedge v_-$ gives the complementary-degree CE pairing and fixes

$$(v_-, v_-)_{\mathrm{gh}} = 1. \tag{22}$$

Unimodularity of $\mathfrak{g}$ makes d graded skew for this pairing. The matter restriction on (21) is δ_{rs}.

Theorem 4.2 (Complete selected residual cohomology)**.** *For the selected derived reduction and centered polarization,*

$$\boxed{H^4_{\mathrm{res}} = \operatorname{span}\{[W_+^2], [W_-^2]\}, \qquad \operatorname{sig} G_{\mathrm{res}} = (2,0), \qquad G_{\mathrm{res}} = I_2.} \tag{23}$$

There is no one-particle class and no higher-matter-weight class.

Proof. Theorem 4.1 removes every noncentered degree. The ghost floor leaves only $N = 0, 1, 2$. The vacuum result, (19), and (20) exhaust them. Equation (22) and the normalized matter restriction give a positive-definite form of signature $(2, 0)$; I_2 is its matrix only after the stated chiral-basis, orientation, and unit-overlap choices. □

5 Krein–Fock completion

Let $\mathcal{W}_{\mathrm{alg}} = \bigoplus_{n\geq 2}^{\mathrm{alg}} \mathcal{H}_n$ and declare its normalized cylinder basis orthonormal for a positive Hilbert majorant. Completing gives $\mathscr{H}_1$. The bounded involution

$$\mathfrak{J}_1 = +1_E \oplus (-1_A) \oplus (-1_L) \tag{24}$$

extends (4); both eigenspaces are infinite-dimensional, so $\mathscr{H}_1$ is a Krein space, not a Pontryagin space. Set

$$\mathscr{F} = \Gamma_s(\mathscr{H}_1), \qquad \mathfrak{J}_{\mathscr{F}} = \Gamma_s(\mathfrak{J}_1), \qquad \mathscr{K} = \mathscr{F} \widehat{\otimes} \Lambda^\bullet \mathfrak{g}^*. \tag{25}$$

Theorem 5.1 (Energy-mode Krein completion)**.** *The residual differential has a closed densely defined maximal extension* $\overline{Q}$ *on* $\mathscr{K}$*, with the algebraic state space as a graph core. Every total-degree eigenspace* $\mathscr{K}_\delta$ *is finite-dimensional. On its noncentered complement the bounded operator*

$$h = \bigoplus_{\delta \neq 0} \frac{\iota_D}{\delta} \tag{26}$$

preserves $\operatorname{Dom} \overline{Q}$ *and satisfies*

$$\overline{Q} h + h \overline{Q} = 1 - P_0. \tag{27}$$

The range of $\overline{Q}$ *is closed, ordinary and reduced cohomology agree, and*

$$\boxed{H^4(\mathscr{K}, \overline{Q}) = \operatorname{span}\{[W_+^2], [W_-^2]\}, \qquad \operatorname{sig} G_{\text{completed}} = (2,0), \qquad G_{\text{completed}} = I_2.} \tag{28}$$

Proof. The ghost compact spectrum lies in $[-4, 4]$. At fixed total degree, matter energy is therefore confined to a finite interval, particle number is bounded by half that energy, and only finitely many finite-dimensional one-particle blocks and bosonic partitions occur. Hence each $\mathscr{K}_\delta$ is finite-dimensional and $\mathscr{K}_0$ is exactly the algebraic centered block.

Writing Q_δ for its finite nilpotent matrix, the maximal Hilbert direct sum with graph domain is closed, and total-degree truncations are a graph core. Since nonzero δ is integral and $\|\iota_D\| = 1$, one has $\|h\| \leq 1$. The block identity and estimate

$$\|Q_\delta h_\delta \psi\| \leq \|\psi\| + \|h_\delta\| \, \|Q_\delta \psi\|$$

give domain invariance and the displayed contraction. The off-center range is a closed kernel, while the centered range is finite-dimensional. Thus the total range is closed and Theorem 4.2 applies without a limiting correction. □

The completion does not exponentiate the unbounded conformal generators or construct a metric-field Cauchy topology. It also does not turn J_{conf} into a positive one-particle form.

6 Descent and the dynamical quotient

For a scalar conformal primary V_h, let

$$\omega = \frac{1}{4!} \epsilon_{\mu\nu\rho\sigma} c^\mu c^\nu c^\rho c^\sigma. \tag{29}$$

The BRST transformations give the same closure defect for ωV_h and the integrated density, proportional to $h/4 - 1$. Hence weight four gives the residual descent

$$[\omega V_4] \longleftrightarrow \left[\int_M \sqrt{-g}\, V_4 \right]. \tag{30}$$

Parity exchanges the chiral classes. In the orthonormal basis

$$e = \frac{W_+^2 + W_-^2}{\sqrt{2}}, \qquad o = \frac{W_+^2 - W_-^2}{\sqrt{2}}, \tag{31}$$

they descend, up to Lorentzian orientation conventions, to C^2 and $C\widetilde{C}$. Their Gram matrix remains I_2.

The Euler–Lagrange map sends e to a nonzero multiple of the Bach tensor and sends o to zero by Chern–Weil transgression. Therefore

$$0 \longrightarrow \mathrm{span}\{o\} \longrightarrow H^4_{\mathrm{res}} \longrightarrow \mathrm{span}\{B_{\mu\nu}\} \longrightarrow 0, \qquad H^4_{\mathrm{dyn}} \cong \mathrm{span}\{[C^2]\}, \quad G_{\mathrm{dyn}} = I_1. \tag{32}$$

The odd class is locally variationally trivial, not globally irrelevant for all boundary conditions or topologies.

7 Focused claim ledger

This page records the claims used by this article and their exact dependency boundary. Detailed sparse matrices, hashes, elapsed times, and generated bases are retained in the companion computational supplement.

Claim	Status/category	Exact receipt or falsification command
Parity-complete $E/A/L$ module, global curvature bridge, and invariant Krein form	Exact; `REDUCED-MODE`	`symbolic/verify_conformal_weyl_module.py`; `symbolic/verify_conformal_cylinder_preimages.py`
Taub/moment-map equality, endpoint duality, and normalized BV–BFV suspension	Exact; `LOCAL-ALGEBRAIC` / `REDUCED-MODE`	`symbolic/verify_conformal_taub_moment_map_all_levels.py`; `symbolic/verify_conformal_taub_obstruction_map.py`; `symbolic/verify_conformal_dual_zero_modes.py`; `symbolic/verify_conformal_residual_bfv_roles.py`; `symbolic/verify_conformal_closed_universe_bfv.py`; `symbolic/verify_conformal_zero_mode_transgression.py`
Centered absolute residual ranks and $H^4_{\mathrm{res}} \cong \mathbb{C}^2$	Exact; `REDUCED-MODE`	run with `python3`: `symbolic/verify_conformal_paper_free.py`; `symbolic/verify_conformal_polarized_state_complex.py`; independent central rank: `symbolic/verify_conformal_residual_rank53_independent.py`
Residual Hermitian signature $(2, 0)$ and normalized I_2	Exact for the selected vertex/deformation basis; `REDUCED-MODE`	`symbolic/verify_conformal_polarized_pairing_transfer.py`; positivity is not a one-particle or graviton-Hilbert claim
Closed maximal residual differential and Krein–Fock cohomology	Exact; `REDUCED-MODE` completion	`symbolic/verify_conformal_energy_mode_krein.py`; no integrated $SO(4, 2)$ representation or metric Cauchy completion claimed
Derived quotient interpretation	Exact in selected formal infinitesimal category; `LOCAL-ALGEBRAIC`	Theorem 3.3; `symbolic/verify_conformal_closed_universe_bfv.py`; no global derived stack, boundary-charge, or fixed-cylinder theorem claimed

Tier boundary. This paper is a residual algebraic and energy-mode completion theorem. It does not use reduced-mode or Euclidean evidence to claim a Lorentzian off-shell BV propagator. The separate causal-curvature paper supplies that bridge and imports Theorem 4.2; it does not recompute the CE group.

8 Conclusion

For the selected closed-universe residual gauging, the Taub constraints are the components of the residual moment map, endpoint duality supplies their BFV cotangent variables, and the Koszul–

CE charge realizes the formal derived zero-locus and stabilizer quotient. Cartan contraction reduces the absolute complex to a finite centered calculation. Exactly two ghost-dressed Weyl-square deformation classes survive, with positive-definite signature $(2, 0)$ and matrix I_2 after the specified normalization. The result survives the selected Krein–Fock completion because every total-degree block is finite and the Cartan homotopy extends boundedly.

This conclusion is deliberately conditional on full residual gauging. A fixed cylinder with D as Hamiltonian, a boundary-charge problem, an interacting theory, or a positive graviton Hilbert-space question requires a different complex and is not answered here.

References

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